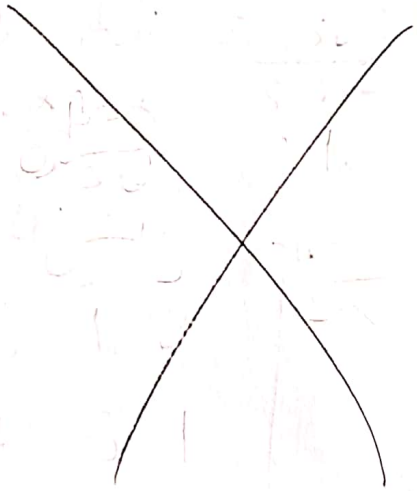
On my honor, I have neither nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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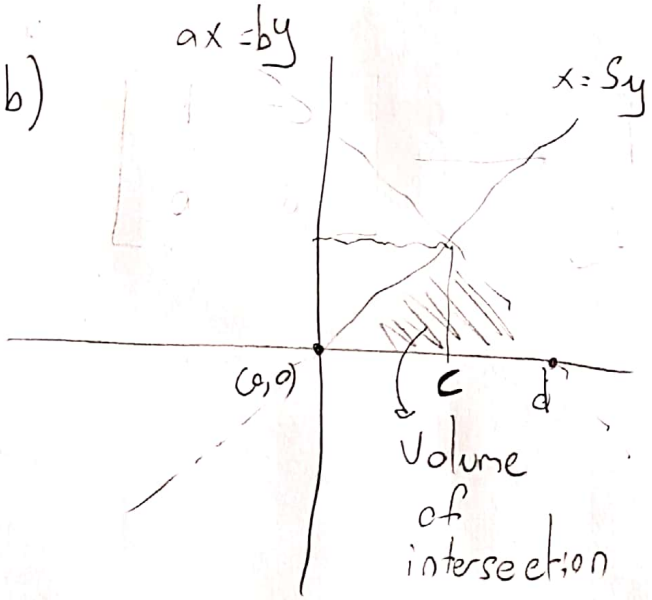
1) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ $\begin{matrix} R_2 = 9R_1 \\ \longrightarrow \\ R_3 - 3R_1 \\ R_4 - 2R_1 \end{matrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$



2) $x = 5y \rightarrow x - 5y = 0$

a) $\underbrace{\begin{bmatrix} 1 & -5 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{B} = 0$ ⇒ Every column at A represents dimensions of equation so the minimum dimention of $x = 5y$ and another equation gives me 2 dimention.

b)



$$\underbrace{\begin{bmatrix} 1 & -5 \\ a & -b \end{bmatrix}}_{A} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = 0$$

$y = \frac{ax}{b}$ $\quad \frac{a}{b} = \frac{x}{5}$

$y = \frac{x}{5}$ $\quad \frac{a}{b} = \frac{1}{5}$

Solution matrix gives me the points and for example like triangular I have to integrate for bases like $\int_0^c \left(\frac{x}{5} - \frac{ax}{b}\right) dx + \int_c^d \left(\frac{ax}{b} - \frac{x}{5}\right) dx$

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c) $x = 5y$

$x + 4y - z = 6$

$$\left[\begin{array}{ccc|c} 1 & -5 & 0 & 0 \\ 1 & 4 & -1 & 6 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{ccc|c} 1 & -5 & 0 & 0 \\ 0 & 9 & -1 & 6 \end{array}\right]$$

$x - 5y = 0 \Rightarrow x = 5y = \frac{30 + 5z}{9}$

$9y - z = 6 \Rightarrow y = \frac{6 + z}{9}$

$z = q$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \left\langle c_1 \begin{bmatrix} 5/9 \\ 1/9 \\ 1 \end{bmatrix} + \begin{bmatrix} 10/3 \\ 2/3 \\ 0 \end{bmatrix}, q \in R \right\rangle$$

From that equation we can say there is a dependency so these vectors (solution vector) is 2D and the intersection is $R^3 \rightarrow R^2$

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3) $H = I - 2uu^T$

a) $H^2 = I$

u : unit → length 1

$u \cdot u^T = I$

Ex:

$$\begin{bmatrix} \cos \\ \sin \end{bmatrix} \begin{bmatrix} \cos & \sin \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$H = -I \Rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$H^2 = I^2 \Rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$H^2 = I$

b) Orthonormal means orthogonal unit matrix and I is a unit vector.

Another expectation is orthogonality means perpendecular so $\det(H) = 1$

$\det(H) = -1 - 0 = -1$

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(3) c) $\lambda = \begin{bmatrix} \cos \\ \sin \end{bmatrix}$

$Hv = \lambda v \Rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} v_1 \end{bmatrix} = \begin{bmatrix} \cos \\ \sin \end{bmatrix} \begin{bmatrix} v_1 \end{bmatrix}$

$\begin{bmatrix} v_1 \end{bmatrix} = \begin{bmatrix} v_2 \end{bmatrix}$

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③ a-) $H=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$

Diagonalization has two properties

1) $T(u+v) = T(u) + T(v)$ ✓

2) $c\,T(u) = T(cu)$ ✓

$$\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} \Rightarrow (-x,-y)$$

$x_1 = u$, $y_1 = v$ $\Rightarrow$ $T(x_1+y_1) = (-x_1,-y_1)$

$T(x_1) + T(y_1) = (-x_1,-y_1)$ $\Rightarrow$ equal

$u = (x_1, y_1)$

$c\cdot T(x_1,y_1) = c\cdot(-x_1,-y_1)$

$T(cx_1, cy_1) = (c\cdot -x_1, c\cdot -y_1)$ $\Rightarrow$ equal

H is diagonalizable

4) $V - w = V_k - w_k$

$$\begin{bmatrix} V_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} V_k \\ w_k \end{bmatrix}$$

a) According to this equation we can say the 2nd equation expresses the process.

b) $V = 10 + V_k . t$

$w = 40 - w_k . t$

$V = 10 + (V-w).t \Rightarrow V = \frac{10 - wt}{(1-t)}$

$w = 40 - (w-V).t \Rightarrow w = \frac{40 + Vt}{(1+t)}$

The tank1 in 1 sec 10 liter liquid 10 liter water so the percantage of liquids is liquid → %50 / water %50

c) For a basic solution in steady state sitation V_k must be equal w_k so;

$V - w = w - V \Rightarrow V = w \Rightarrow$ 50 total liquids must be equal in two tanks 25 = 25

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